

Capture the Flag

Your Student Portal is now dynamic thanks to interpolation.

But there's a problem.

Your announcement component currently has something like:

```
announcementTitle1 = "Enrollment is Now Open";  
announcementBody1 = "Enrollment for the next semester is now  
available.";
```

```
announcementTitle2 = "Intramurals 2026";  
announcementBody2 = "Registration is now open.";
```

```
announcementTitle3 = "Library Schedule";  
announcementBody3 = "The library will be open until 8 PM.";
```

And your HTML has to repeat the same card structure three times.

The school now has **20 announcements**.

Your mission:

Stop repeating yourself. Make Angular generate the cards for you.

Flag 0 – Mission Branch

Create and switch to a new branch: `feature/ngFor`

Capture

- Show your current Git branch.



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
PS C:\Users\nielj\my-first-angular-app> git status  
On branch feature/interpolation  
nothing to commit, working tree clean  
PS C:\Users\nielj\my-first-angular-app> git branch "feature/ngFor"  
PS C:\Users\nielj\my-first-angular-app>  
PS C:\Users\nielj\my-first-angular-app> git checkout feature/ngFor  
Switched to branch 'feature/ngFor'  
PS C:\Users\nielj\my-first-angular-app> |
```

-

FLAG 1 — The Repetition Problem

Open your announcement component.

Look at your existing announcement cards.

You should currently have something similar to:

```
<div class="announcement-card">
  <h3>{{ announcementTitle1 }}</h3>
  <p>{{ announcementBody1 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle2 }}</h3>
  <p>{{ announcementBody2 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle3 }}</h3>
  <p>{{ announcementBody3 }}</p>
</div>
```

Mission

Imagine the school gives you **10 more announcements**.

Would you want to copy and paste the card **10 more times**?

Don't change anything yet.

📸 **Capture:** Your repeated announcement cards.

TS profile-card.ts

announcement-card.html X

src > app > components > announcement-card > announcement-card.html > ...

Go to component

```
1  <div class="announcement-card">
2    <h3>{{ announcementTitle1 }}</h3>
3    <p>{{ announcementBody1 }}</p>
4    <small>{{ announcementDate1 }}</small>
5  </div>
6
7  <div class="announcement-card">
8    <h3>{{ announcementTitle2 }}</h3>
9    <p>{{ announcementBody2 }}</p>
10   <small>{{ announcementDate2 }}</small>
11 </div>
12
13 <div class="announcement-card">
14   <h3>{{ announcementTitle3 }}</h3>
15   <p>{{ announcementBody3 }}</p>
16   <small>{{ announcementDate3 }}</small>
17 </div>
18
19 <div class="announcement-card">
20   <h3>{{ announcementTitle4 }}</h3>
21   <p>{{ announcementBody4 }}</p>
22   <small>{{ announcementDate4 }}</small>
23 </div>
24
25 <div class="announcement-card">
26   <h3>{{ announcementTitle5 }}</h3>
27   <p>{{ announcementBody5 }}</p>
28   <small>{{ announcementDate5 }}</small>
29 </div>
30
31 <div class="announcement-card">
32   <h3>{{ announcementTitle6 }}</h3>
33   <p>{{ announcementBody6 }}</p>
34   <small>{{ announcementDate6 }}</small>
35 </div>
36
```

src > app > components > announcement-card > <> announcement-card.html > ...

```
30
31 <div class="announcement-card">
32   <h3>{{ announcementTitle6 }}</h3>
33   <p>{{ announcementBody6 }}</p>
34   <small>{{ announcementDate6 }}</small>
35 </div>
36
37 <div class="announcement-card">
38   <h3>{{ announcementTitle7 }}</h3>
39   <p>{{ announcementBody7 }}</p>
40   <small>{{ announcementDate7 }}</small>
41 </div>
42
43 <div class="announcement-card">
44   <h3>{{ announcementTitle8 }}</h3>
45   <p>{{ announcementBody8 }}</p>
46   <small>{{ announcementDate8 }}</small>
47 </div>
48
49 <div class="announcement-card">
50   <h3>{{ announcementTitle9 }}</h3>
51   <p>{{ announcementBody9 }}</p>
52   <small>{{ announcementDate9 }}</small>
53 </div>
54
55 <div class="announcement-card">
56   <h3>{{ announcementTitle10 }}</h3>
57   <p>{{ announcementBody10 }}</p>
58   <small>{{ announcementDate10 }}</small>
59 </div>
60
61
```

FLAG 2 — Put the Data Together

Instead of creating separate variables, create **one array**.

Research Hint

An array can contain multiple objects.

You don't need to memorize the syntax.

Search:

Angular array of objects

 Capture: Your `announcements` array.

```
src > profile-card.ts TS announcement-card.html 3. M TS announcement-card.ts M # announcement-card.css
src > app > components > announcement-card > TS announcement-card.ts > AnnouncementCard > announcements
1 import { Component } from '@angular/core';
2 import { IsActive } from '@angular/router';
3
4
5 @component({
6 selector: 'app-announcement-card',
7 imports: [],
8 templateUrl: './announcement-card.html',
9 styleUrls: ['./announcement-card.css'],
10 })
11 export class AnnouncementCard {
12
13   announcements: { title: string; body: string; date: string; isActive: boolean }[] = [
14     { title: 'Enrollment Schedule', body: 'Enrollment begins next Monday. Please prepare all required documents.',
15       date: 'August 1, 2026', isActive: true },
16     { title: 'Canteen Menu Update', body: 'The canteen menu has been updated for the new semester. Check out the latest options available.',
17       date: 'August 4, 2026', isActive: false },
18     { title: 'Valorant Tournament', body: 'The Valorant tournament will be held next month. Sign up now to participate and showcase your skills!',
19       date: 'August 10, 2026', isActive: true },
20     { title: 'Library Renovation', body: 'The library will be closed for renovation from August 15 to September 15. Please plan your visits accordingly.',
21       date: 'August 12, 2026', isActive: false },
22     { title: 'New Course Offering', body: 'We are excited to announce a new course offering in Artificial Intelligence starting this semester.',
23       date: 'August 20, 2026', isActive: true },
24     { title: 'Career Fair', body: 'The annual career fair will be held on September 5. Meet potential employers and explore job opportunities.',
25       date: 'August 25, 2026', isActive: false },
26     { title: 'Sports Day', body: 'Join us for the annual sports day event on September 10. Participate in various sports activities and cheer for your favorite teams.',
27       date: 'August 30, 2026', isActive: true },
28     { title: 'Guest Lecture Series', body: 'We are hosting a guest lecture series featuring industry experts. Attend these sessions to gain insights and knowledge from professionals.',
29       date: 'September 1, 2026', isActive: false },
30     { title: 'Community Service Initiative', body: 'Get involved in our community service initiative. Volunteer for various projects and make a positive impact in the community.',
31       date: 'September 5, 2026', isActive: true },
32     { title: 'End-of-Semester Celebration', body: 'Join us for the end-of-semester celebration on September 15. Celebrate your achievements and enjoy a fun-filled event with your peers.',
33       date: 'September 10, 2026', isActive: false },
34     { title: 'Alumni Meet', body: 'Reconnect with your fellow alumni at the upcoming alumni meet. Share experiences, network, and reminisce about your time at the university.',
35       date: 'September 20, 2026', isActive: true },
36     { title: 'Graduation Ceremony', body: 'Mark your calendars for the upcoming graduation ceremony on September 25. Celebrate the achievements of our graduating class.',
37       date: 'September 15, 2026', isActive: false }
38   ];
39 }
```

FLAG 3 — Meet **ngFor*

Now comes the important part.

You have:

```
announcements = [  
  { ... },  
  { ... },  
  { ... }  
];
```

How can Angular create one card **for every item** in that array?

Hint

Research:

**Angular ngFor*

You should discover syntax similar to:

```
*ngFor="let announcement of announcements"
```

Your mission is to use it on your announcement card.

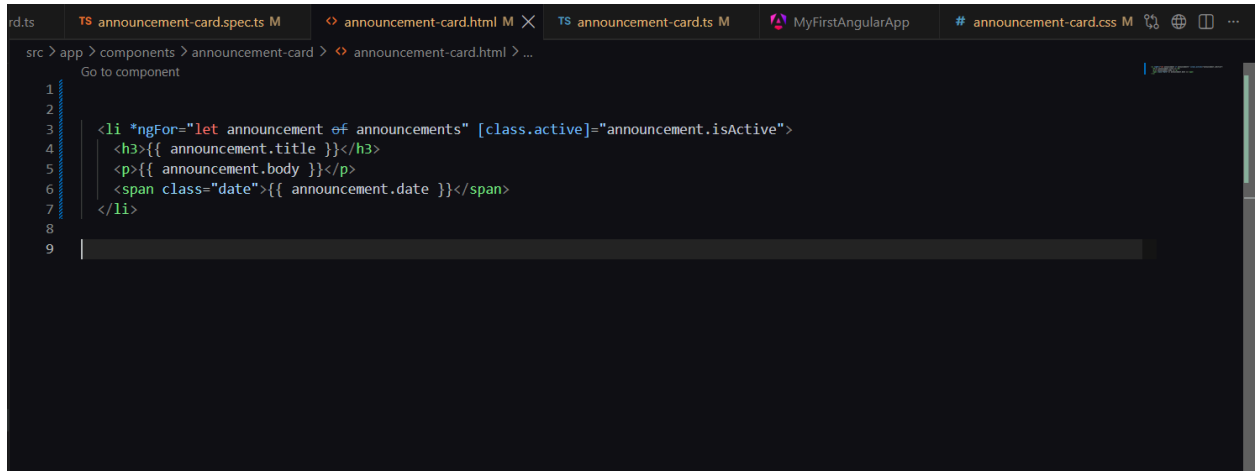
Instead of having:

```
<div class="announcement-card">
```

repeated three times, you should have **one card template** that Angular repeats.

 **Capture:**

- Your updated code.

A screenshot of a code editor window showing an Angular component template. The editor has several tabs at the top: 'rd.ts', 'TS announcement-card.spec.ts M', 'announcement-card.html M', 'TS announcement-card.ts M', 'MyFirstAngularApp', and 'announcement-card.css M'. The active tab is 'announcement-card.html M'. The code in the editor is as follows:

```
1 <li *ngFor="let announcement of announcements" [class.active]="announcement.isActive">
2
3   <h3>{{ announcement.title }}</h3>
4   <p>{{ announcement.body }}</p>
5   <span class="date">{{ announcement.date }}</span>
6 </li>
7
8
9
```

FLAG 4 — Display the Current Announcement

You now have a loop variable.

Inside the repeated card, display title and body:

Your browser should now display all eight announcements.

Important

You should have:

1 array

1 card template

8 displayed cards

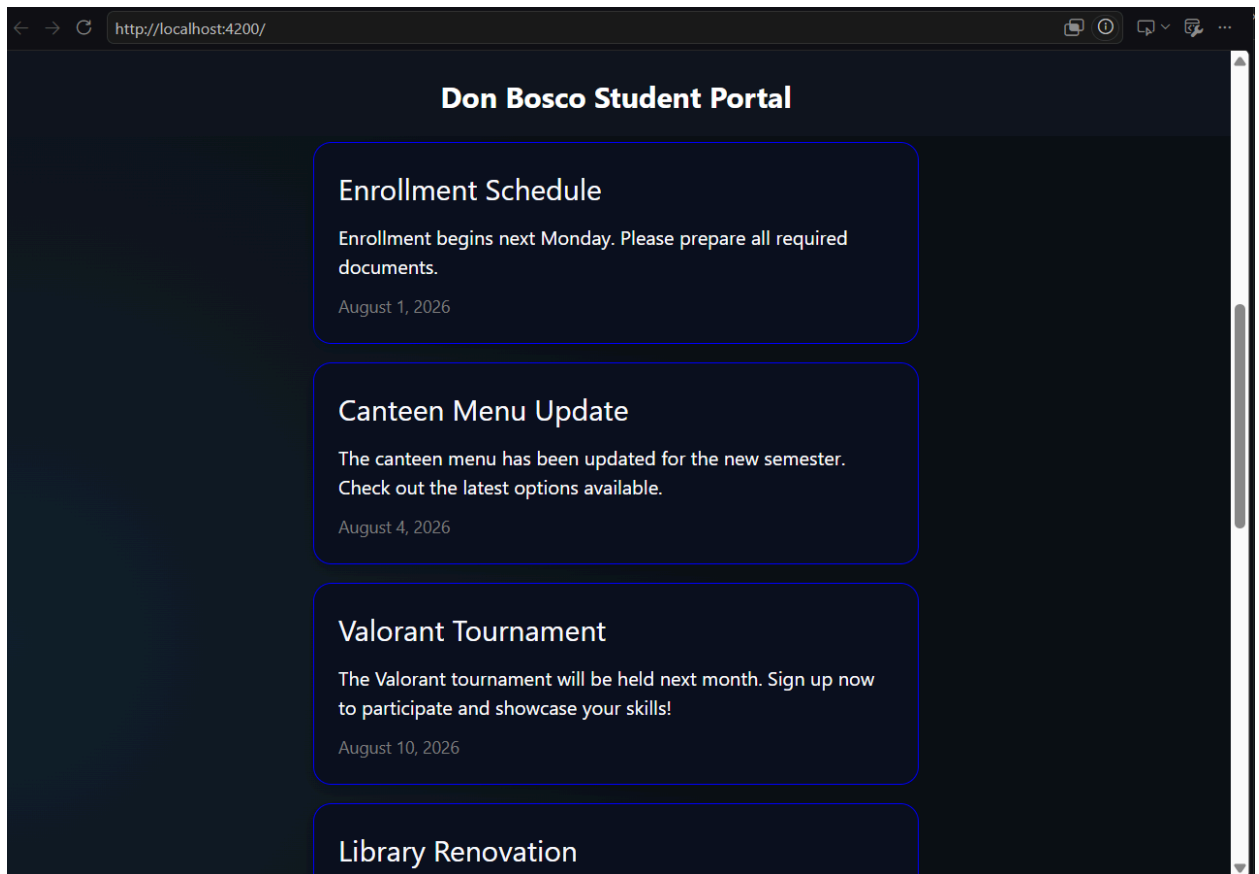
Not:

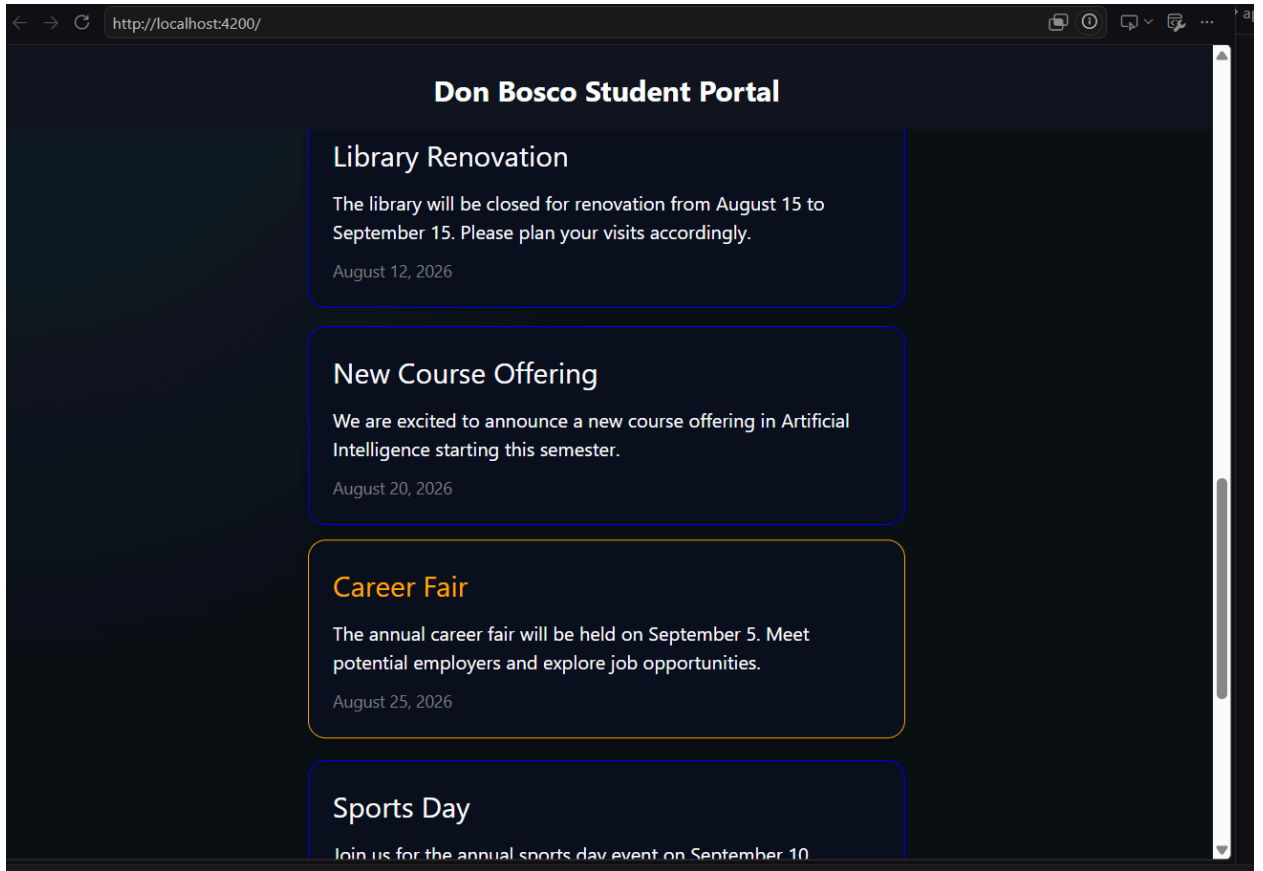
8 arrays

8 card templates

 **Capture:**

- Browser showing all announcements.





- Updated code.

```
src > app > components > announcement-card > TS announcement-card.ts > AnnouncementCard
6   selector: 'app-announcement-card',
7   imports: [CommonModule],
8   templateUrl: './announcement-card.html',
9   styleUrls: ['./announcement-card.css'],
10  })
11  export class AnnouncementCard {
12
13    announcements: { title: string; body: string; date: string; isActive: boolean }[] = [
14      { title: 'Enrollment Schedule', body: 'Enrollment begins next Monday. Please prepare all required documents.',
15        date: 'August 1, 2026', isActive: true},
16      { title: 'Canteen Menu Update', body: 'The canteen menu has been updated for the new semester. Check out the latest
17        date: 'August 4, 2026', isActive: false},
18      { title: 'Valorant Tournament', body: 'The Valorant tournament will be held next month. Sign up now to participate a
19        date: 'August 10, 2026', isActive: true},
20      { title: 'Library Renovation', body: 'The library will be closed for renovation from August 15 to September 15. Plea
21        date: 'August 12, 2026', isActive: false},
22      { title: 'New Course Offering', body: 'We are excited to announce a new course offering in Artificial Intelligence s
23        date: 'August 20, 2026', isActive: true},
24      { title: 'Career Fair', body: 'The annual career fair will be held on September 5. Meet potential employers and expl
25        date: 'August 25, 2026', isActive: false},
26      { title: 'Sports Day', body: 'Join us for the annual sports day event on September 10. Participate in various sports
27        date: 'August 30, 2026', isActive: true},
28      { title: 'Guest Lecture Series', body: 'We are hosting a guest lecture series featuring industry experts. Attend the
29        date: 'September 1, 2026', isActive: false},
30    ]
31  }
32
33  }
```

```
src > app > components > announcement-card > announcement-card.html > HTML Language Features > div.announcement-card
Go to component
1
2
3  <div
4    class="announcement-card"
5    *ngFor="let announcement of announcements"
6    [class.active]="announcement.isActive"
7  >
8
9    <h3>{{ announcement.title }}</h3>
10   <p>{{ announcement.body }}</p>
11   <span class="date">{{ announcement.date }}</span>
12 </div>
```

FLAG 5 — Add More Data

Now test whether your loop actually works.

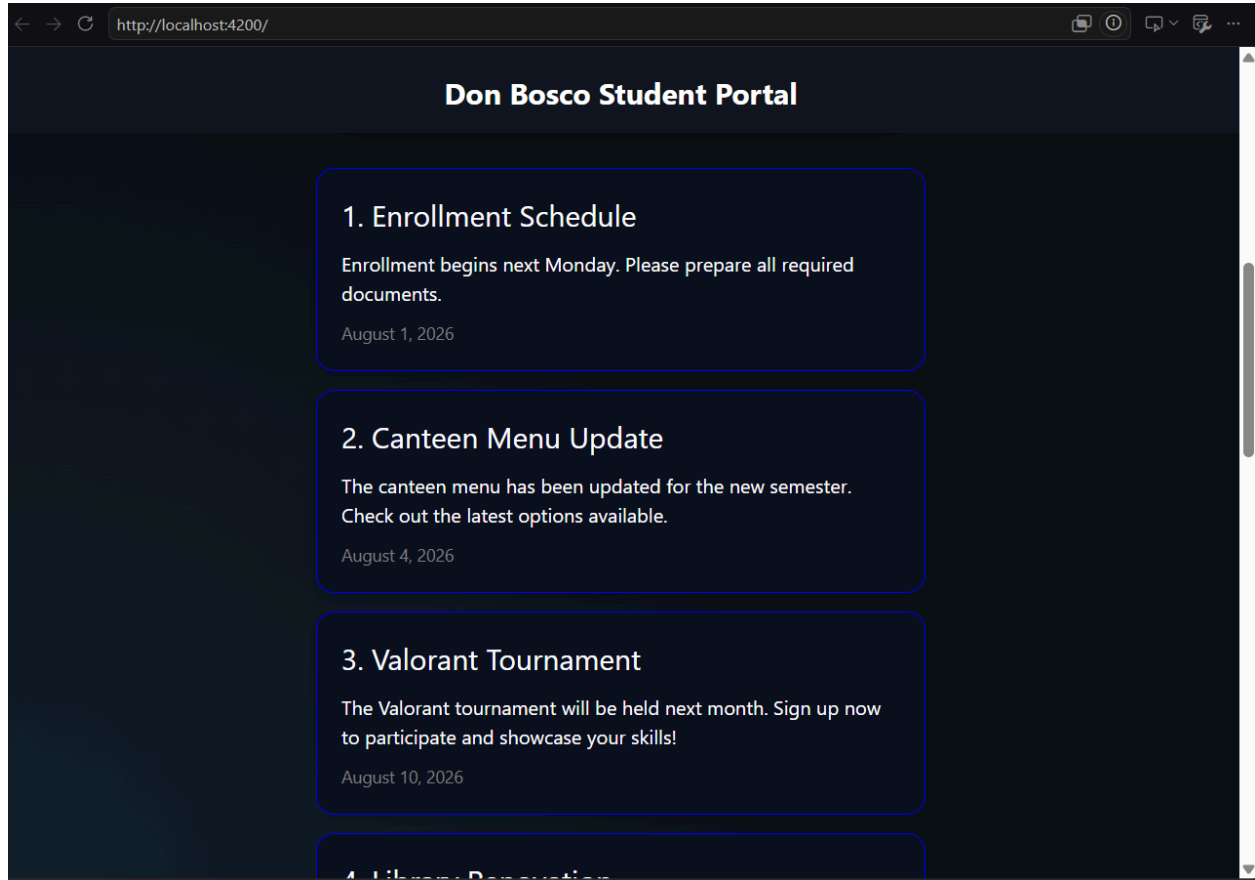
Add **two more announcements** to the array.

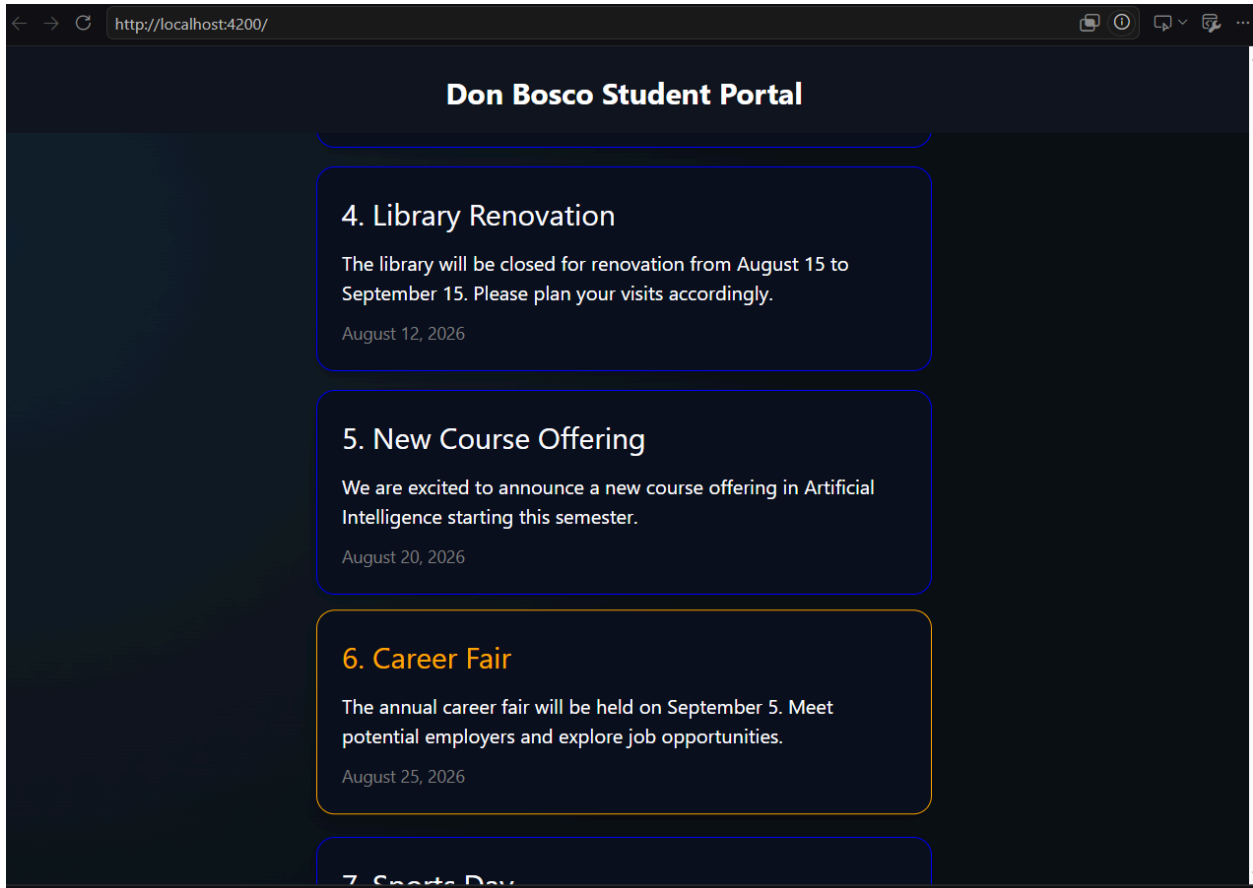
You should **not change your HTML**.

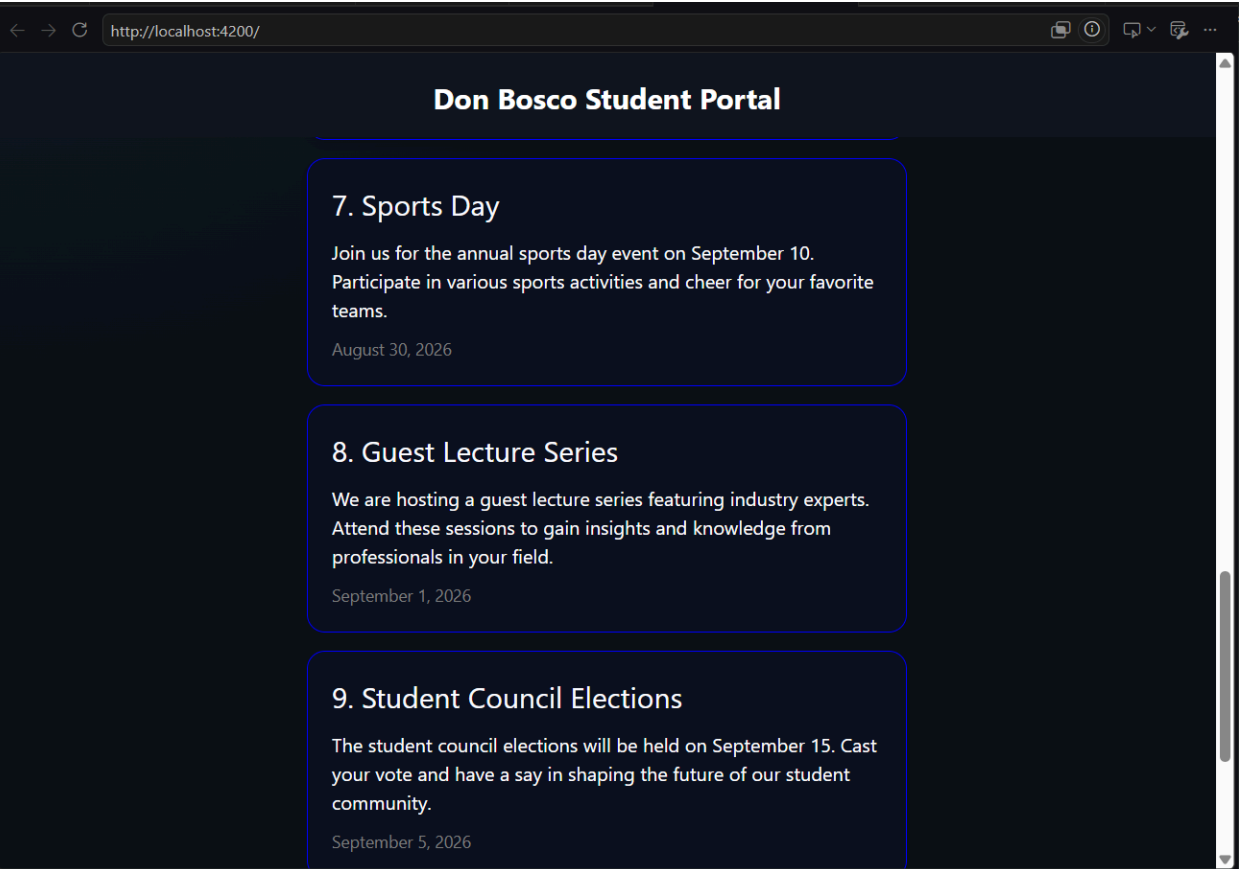
If two additional cards automatically appear:

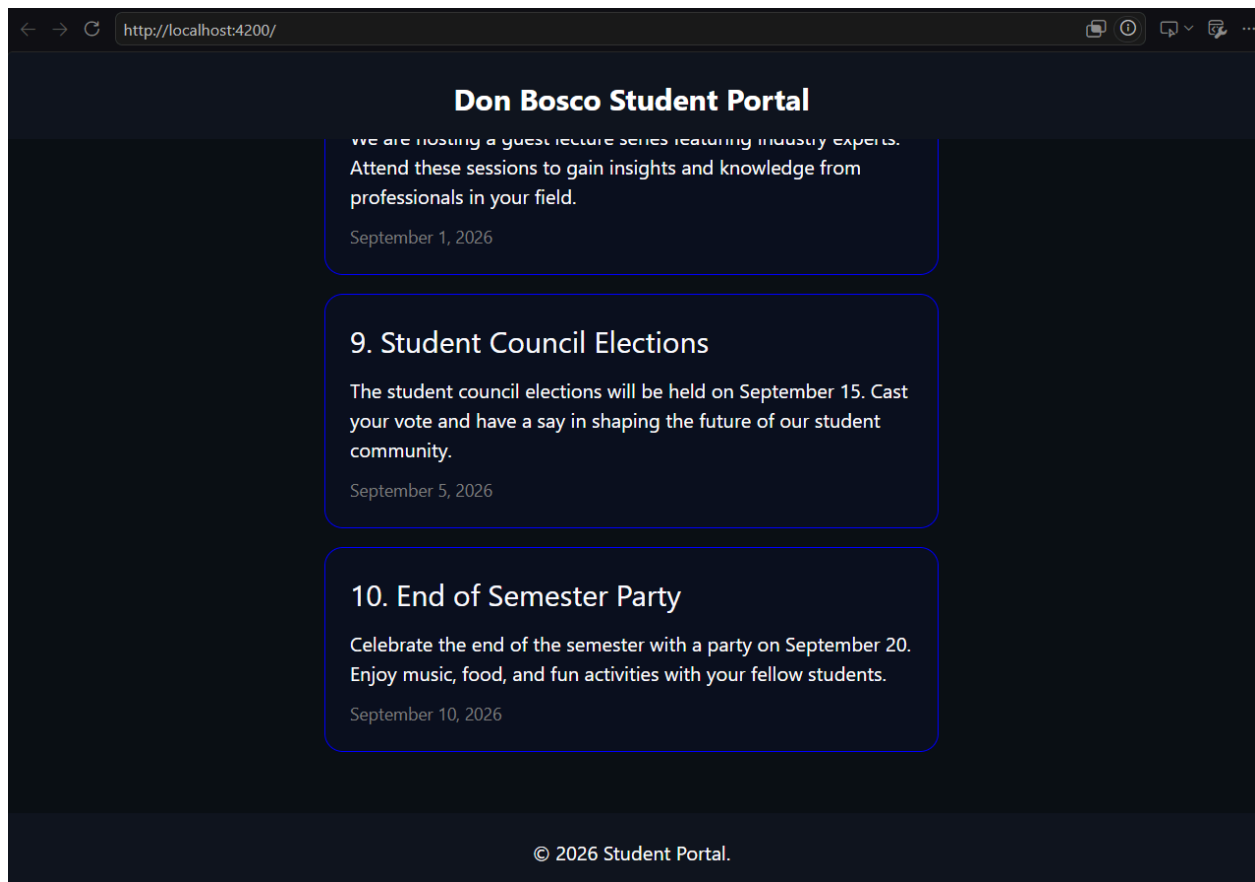
🚩 **FLAG CAPTURED**

 **Capture:** Browser showing all ten announcements.









FLAG 6 — Make the Cards More Realistic

Your announcement objects currently contain only:

title
body

Add more information: author and date.

Update your card to display all fields.

Use **interpolation** for all values.

📸 **Capture:** Your richer announcement cards.

1. Enrollment Schedule

Enrollment begins next Monday. Please prepare all required documents.

By: John Doe
August 1, 2026

2. Canteen Menu Update

The canteen menu has been updated for the new semester. Check out the latest options available.

By: Jane Smith
August 4, 2026

3. Valorant Tournament

The Valorant tournament will be held next month. Sign up now to participate and showcase your skills!

By: John Doe
August 10, 2026

FLAG 7 — Student List

Announcements aren't the only thing that can be repeated.

Create another array named students.

Create a simple student card.

Use `*ngFor` to display every student.

Expected example:

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

 **Capture:** Student list.

Juan Dela Cruz BSIT Year: 2nd Year
Maria Santos Reyes BSCS Year: 3rd Year
Pedro Gonzales Lopez BSIT Year: 1st Year

FLAG 8 — One Template, Many Students

Add another student to your array.

Do **not** modify the HTML.

The new student should automatically appear.

This is the important concept:

DATA CHANGES



ARRAY CHANGES



ANGULAR GENERATES THE UI

 **Capture:** Before and after adding the student.

Juan Dela Cruz BSIT Year: 2nd Year
Maria Santos Reyes BSCS Year: 3rd Year
Pedro Gonzales Lopez BSIT Year: 1st Year

Juan Dela Cruz BSIT Year: 2nd Year
Maria Santos Reyes BSCS Year: 3rd Year
Pedro Gonzales Lopez BSIT Year: 1st Year
Niel Jhon Celocia BSIT Year: 4th Year

FLAG 9 — Research Challenge

I will **not** provide the next syntax.

Research:

**Angular ngFor index*

Find out how to access the position of the current item.

Use it to display:

#1 Juan Dela Cruz

#2 Maria Santos

#3 Pedro Garcia

Without hardcoding the numbers.

Hint:

let i = index

 **Capture:**

- Your numbered student list.

1. Juan Dela Cruz
BSIT
Year: 2nd Year

2. Maria Santos Reyes
BSCS
Year: 3rd Year

3. Pedro Gonzales Lopez
BSIT
Year: 1st Year

4. Niel Jhon Celocia
BSIT
Year: 4th Year

-
- Your code

```
src > app > components > students > < students.html > ...  
Go to component  
1  <div class="student-card"  
2    *ngFor="let student of students; let i = index"  
3    [class.active]="student.isActive"  
4  >  
5    <h3>{{ i + 1 }}. {{ student.firstName }} {{ student.middleName }} {{ student.lastName }}</h3>  
6    <p>{{ student.Course }}</p>  
7    <span class="year-level">Year: {{ student.YearLevel }}</span>  
8  </div>  
9  
10  
11
```

FLAG 10 — Empty List

Now test what happens when there are no announcements.

Change:

```
announcements = [];
```

What happens?

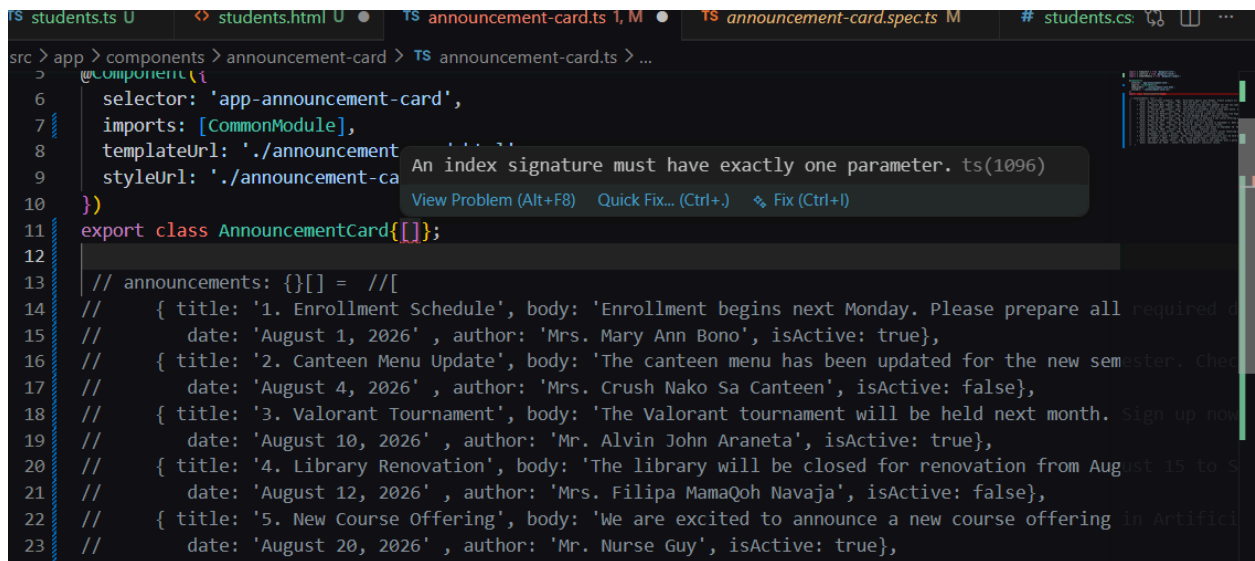
Write a short observation in your documentation:

What happens when the array is empty?

Answer: it asks for data.

Then restore your announcements.

 **Capture:** Your experiment and observation.



```
src > app > components > announcement-card > TS announcement-card.ts > ...
6   selector: 'app-announcement-card',
7   imports: [CommonModule],
8   templateUrl: './announcement-card.component.html',
9   styleUrls: ['./announcement-card.component.css'],
10  })
11  export class AnnouncementCard{[]};
12
13  // announcements: {}[] = //[
14  //   { title: '1. Enrollment Schedule', body: 'Enrollment begins next Monday. Please prepare all required d
15  //     date: 'August 1, 2026' , author: 'Mrs. Mary Ann Bono', isActive: true},
16  //   { title: '2. Canteen Menu Update', body: 'The canteen menu has been updated for the new semester. Chec
17  //     date: 'August 4, 2026' , author: 'Mrs. Crush Nako Sa Canteen', isActive: false},
18  //   { title: '3. Valorant Tournament', body: 'The Valorant tournament will be held next month. Sign up now
19  //     date: 'August 10, 2026' , author: 'Mr. Alvin John Araneta', isActive: true},
20  //   { title: '4. Library Renovation', body: 'The library will be closed for renovation from August 15 to 9
21  //     date: 'August 12, 2026' , author: 'Mrs. Filipa MamaQoh Navaja', isActive: false},
22  //   { title: '5. New Course Offering', body: 'We are excited to announce a new course offering in Artificial
23  //     date: 'August 20, 2026' , author: 'Mr. Nurse Guy', isActive: true},
```

FLAG 11 — Dynamic Student Portal

Your Student Portal should now contain:

Your Student Profile

Still using interpolation:

Name

Student ID
Course
Year
Grade
Academic Standing
Email

Announcements

Generated from:

announcements = [...]

Using:

*ngFor

Student List

Generated from:

students = [...]

Using:

*ngFor and index

Your final requirement

Final Screenshot

- Entire browser UI of all requirements in this activity

Don Bosco Student Portal



NIEL JHON E. CELOCIA

2023-00000

BSIT

4th Year - Valorant

Nickname: pako

Email: niel.jhon.celocia@dbtc-cebu.edu.ph

Grade: 2.5

Status: Pass

Don Bosco Student Portal

Enrollment Schedule

Enrollment begins next Monday. Please prepare all required documents.

by: Mrs. Mary Ann Bono
August 1, 2026

Canteen Menu Update

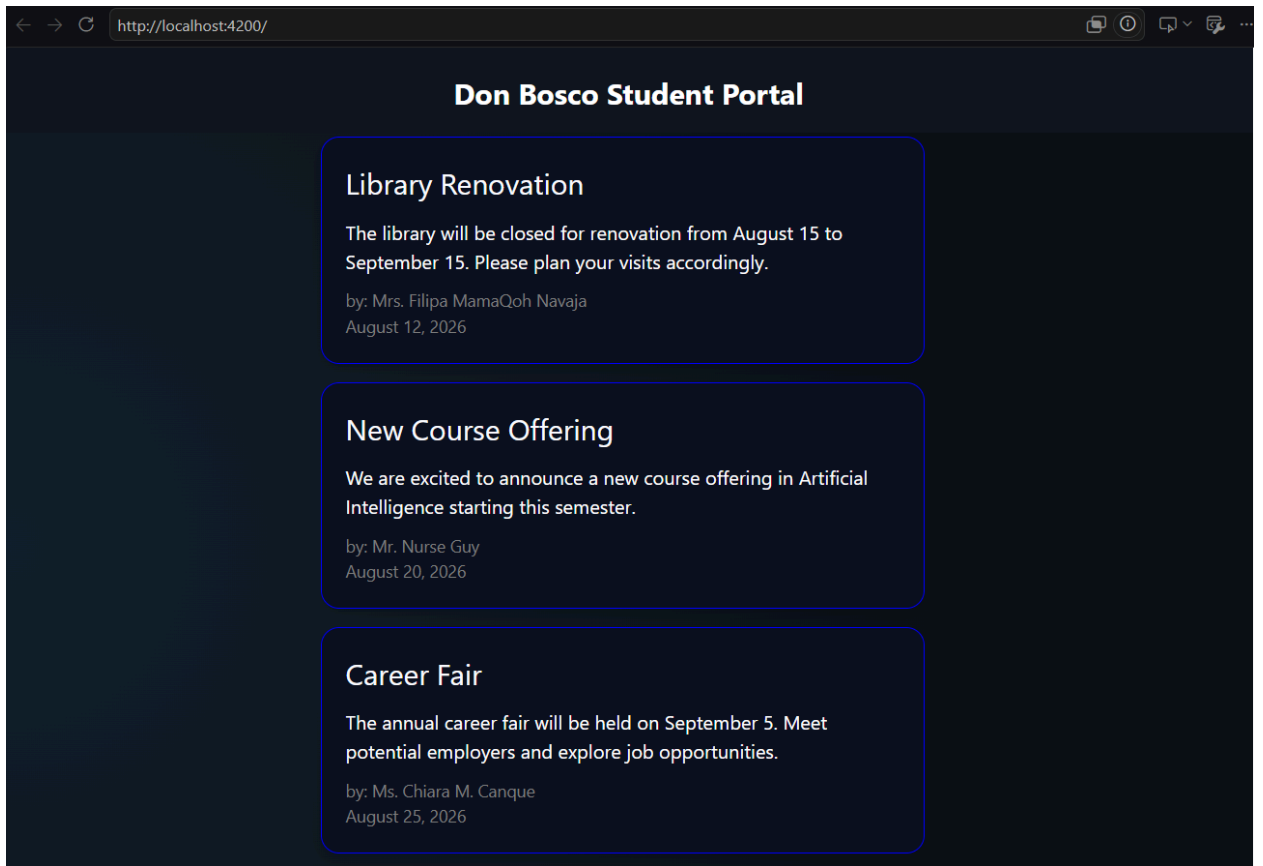
The canteen menu has been updated for the new semester. Check out the latest options available.

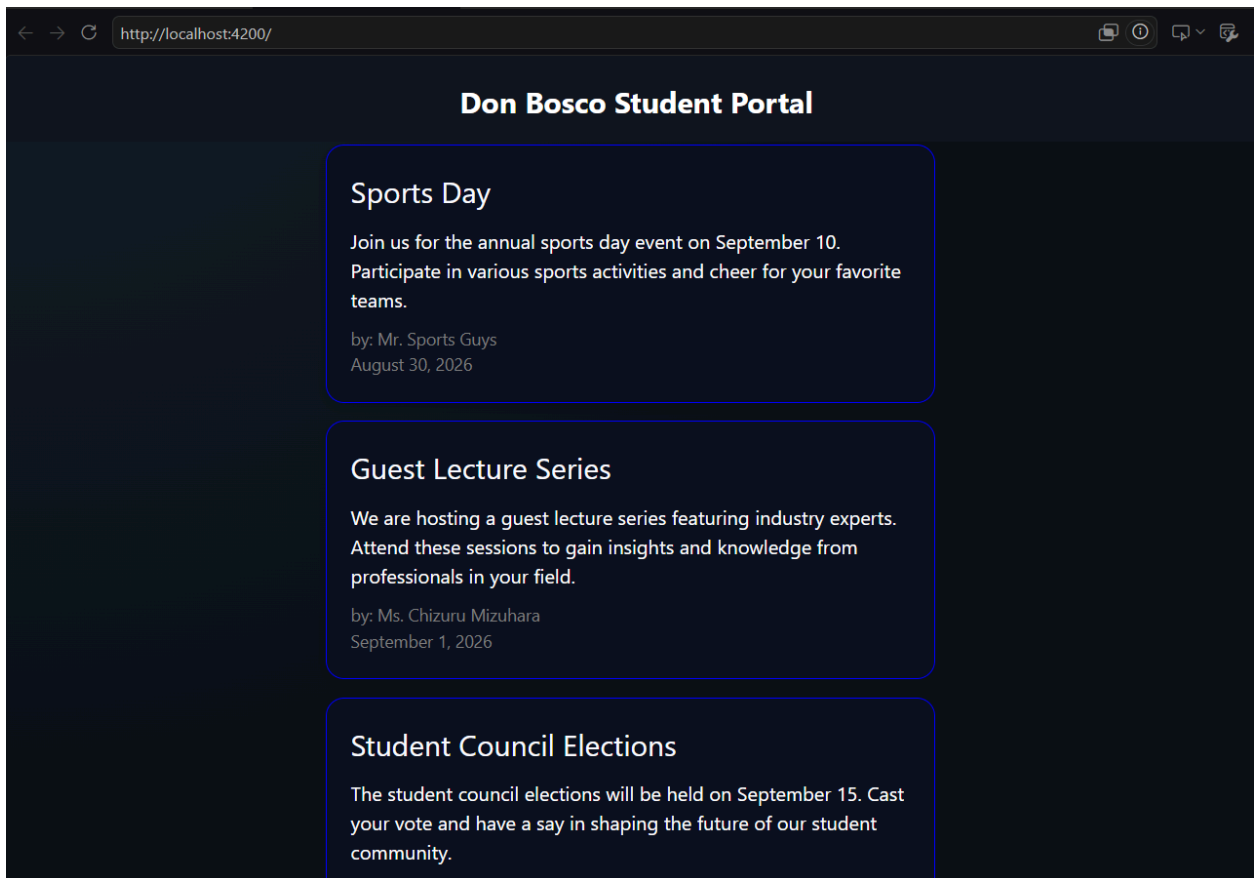
by: Mrs. Crush Nako Sa Canteen
August 4, 2026

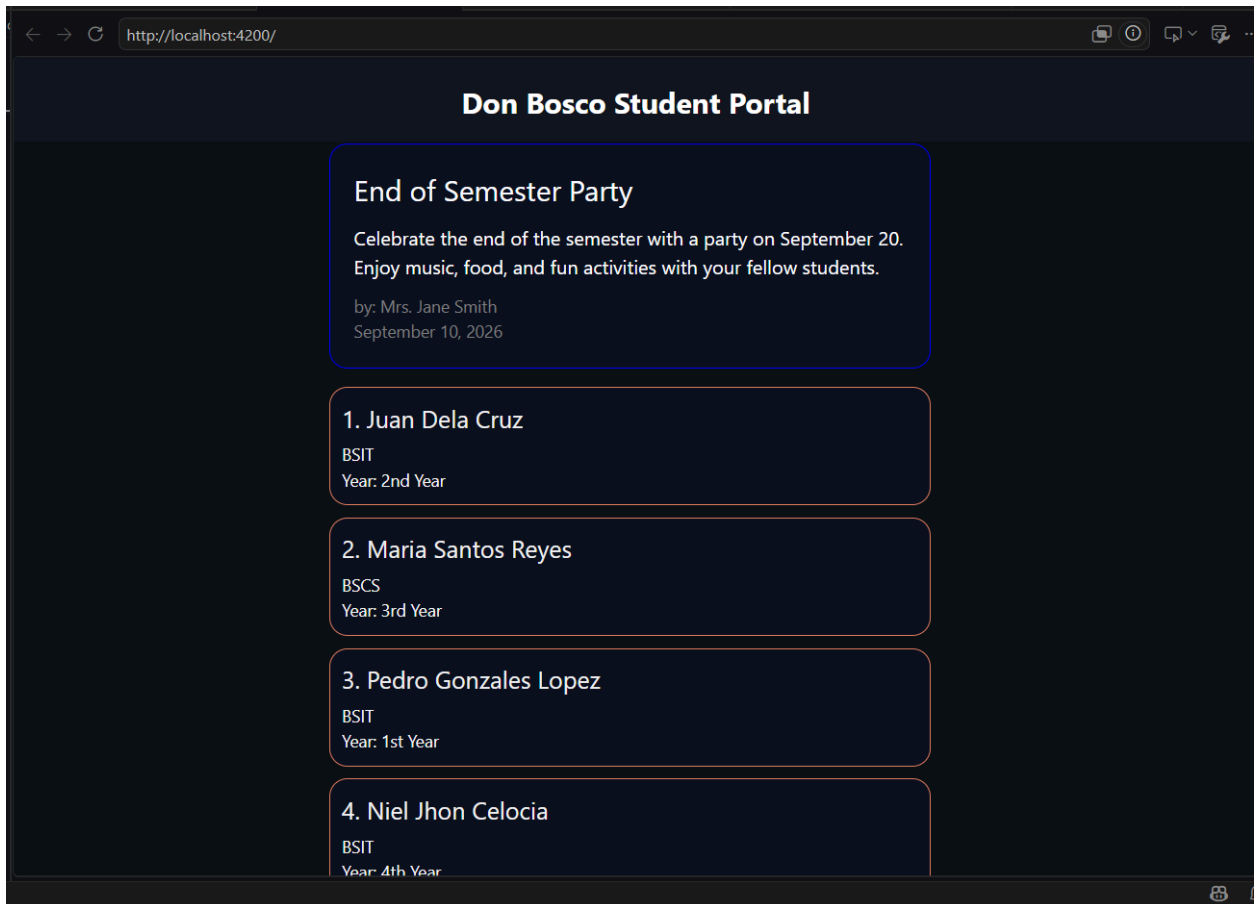
Valorant Tournament

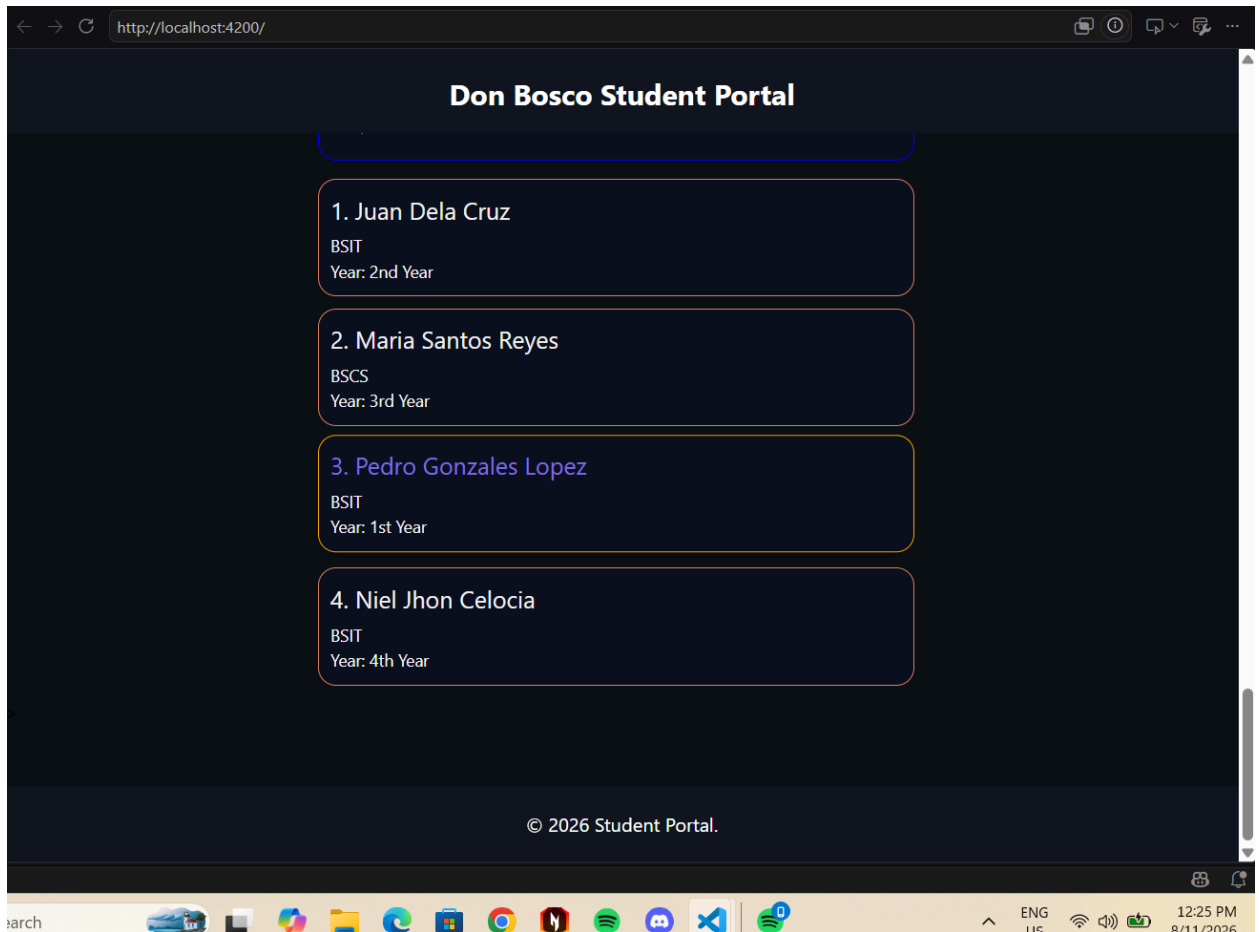
The Valorant tournament will be held next month. Sign up now to participate and showcase your skills!

by: Mr. Alvin John Araneta
August 10, 2026









FLAG 12 — Commit & Push

Final Screenshot

- Successful push to feature/ngFor

